

Case Study Document

Topic: Underfitting, Overfitting, and Good Fit

Case Study Name

EV Battery Warranty Claim Prediction

1. Business Problem

An electric vehicle manufacturer wants to predict whether an EV battery may create a warranty claim in the next 6 months.

Battery replacement is costly, so the company wants to identify risky vehicles early using vehicle usage and battery health data.

2. Prediction Goal

Predict whether a vehicle will have:

- **Warranty Claim**
 - **No Warranty Claim**
-

3. Dataset Description

Feature	Meaning	Example
vehicle_age_months	Age of vehicle in months	28
km_driven	Total distance driven	46000
battery_health_score	Battery health score out of 100	62
charging_cycles	Number of charging cycles	720
fast_charging_percent	Percentage of fast charging usage	65
avg_daily_km	Average daily running	72
temperature_exposure	Low / Medium / High heat exposure	High
fault_code_count	Battery-related fault codes	5

Feature	Meaning	Example
service_visits	Previous service visits	3
driving_style	Smooth / Moderate / Aggressive	Aggressive
target	Actual result	Claim / No Claim

4. Learning Objective

Learners will understand how model complexity affects prediction performance.

Model Behavior	Meaning
Underfitting	Model is too simple
Overfitting	Model is too complex
Good Fit	Model is balanced

5. Algorithm to Use

Use **Decision Tree Classifier**.

Decision Tree is suitable because it creates easy-to-understand rules.

Example rule:

```
IF battery_health_score is low
AND fault_code_count is high
AND fast_charging_percent is high
THEN Warranty Claim
```

Activity Steps

Step 1: Understand the Business Problem

Explain the business requirement:

The EV manufacturer wants to predict future battery warranty claims before the issue becomes expensive.

Discuss why this is important:

Business Benefit	Explanation
Reduce warranty cost	Identify risky battery packs early
Improve customer experience	Prevent sudden battery failure
Better service planning	Prioritize vehicles for inspection
Product improvement	Understand failure patterns

Step 2: Identify Input and Output

Separate the data into input features and target output.

Input Features

```
vehicle_age_months  
km_driven  
battery_health_score  
charging_cycles  
fast_charging_percent  
avg_daily_km  
temperature_exposure  
fault_code_count  
service_visits  
driving_style
```

Target Output

```
target
```

Target value:

```
Claim / No Claim
```

Step 3: Prepare the Data

Convert text columns into numeric format.

Column	Original Values	Required Action
temperature_exposure	Low / Medium / High	Convert into numbers or encoded columns
driving_style	Smooth / Moderate / Aggressive	Convert into numbers or encoded columns
target	Claim / No Claim	Convert into 1 / 0

Reason:

Machine learning models work with numbers.

Step 4: Split the Data

Divide the dataset into:

Dataset	Purpose
Training Data	Used to train the model
Testing Data	Used to test the model on unseen data

Suggested split:

```
70% Training
30% Testing
```

Model 1: Underfitting Model

Step 5: Train a Very Simple Decision Tree

Create a Decision Tree with very low depth.

Suggested setting:

```
max_depth = 1
```

This means the model can ask only one major question.

Example:

Is battery_health_score less than a certain value?

Step 6: Evaluate the Underfitting Model

Check the following:

Metric	Purpose
Training Accuracy	How well the model performs on training data
Testing Accuracy	How well the model performs on new data

Step 7: Learner Observation for Underfitting

Learners should answer:

Question	Observation
Is training accuracy low or high?	To be filled
Is testing accuracy low or high?	To be filled
Is the model too simple?	To be filled
Which important battery factors may be ignored?	To be filled

Model 2: Overfitting Model

Step 8: Train a Very Complex Decision Tree

Create a Decision Tree with very high depth or unlimited depth.

Suggested setting:

max_depth = None

or

```
max_depth = 12
```

This allows the model to create many detailed rules.

Example:

```
IF vehicle_age_months = 28
AND charging_cycles = 720
AND fault_code_count = 5
AND driving_style = Aggressive
THEN Claim
```

This type of rule may become too specific.

Step 9: Evaluate the Overfitting Model

Check the following:

Metric	Purpose
Training Accuracy	To check whether the model memorized training data
Testing Accuracy	To check whether the model works on unseen vehicles

Step 10: Learner Observation for Overfitting

Learners should answer:

Question	Observation
Is training accuracy very high?	To be filled
Is testing accuracy much lower than training accuracy?	To be filled
Is the model memorizing past vehicles?	To be filled
Will this model work well on new EVs?	To be filled

Model 3: Good Fit Model

Step 11: Train a Balanced Decision Tree

Create a Decision Tree with controlled depth.

Suggested setting:

```
max_depth = 3
```

or

```
max_depth = 4
```

The model should learn useful patterns without memorizing every record.

Example balanced pattern:

```
Low battery health  
High fast charging usage  
High charging cycles  
High battery fault codes
```

Step 12: Evaluate the Good Fit Model

Check the following:

Metric	Purpose
Training Accuracy	Should be reasonably good
Testing Accuracy	Should be close to training accuracy

Step 13: Learner Observation for Good Fit

Learners should answer:

Question	Observation
Is training accuracy good?	To be filled
Is testing accuracy close to training accuracy?	To be filled
Is the model balanced?	To be filled
Can this model help the warranty team?	To be filled

Comparison Activity

Step 14: Compare All Three Models

Learners should fill the table after running the models.

Model Type	Tree Depth	Training Accuracy	Testing Accuracy	Behavior
Underfitting Model	1	To be filled	To be filled	To be identified
Good Fit Model	3 or 4	To be filled	To be filled	To be identified
Overfitting Model	None / High	To be filled	To be filled	To be identified

Step 15: Business Discussion Questions

Question
Which model may miss risky batteries?
Which model may create false warranty alerts?
Which model may memorize old EV battery records?
Which model is better for predicting new vehicles?
What happens if a risky battery is predicted as No Claim?
What happens if a healthy battery is predicted as Claim?

Step 16: Learner Deliverables

Learners should submit:

Deliverable	Description
Data preparation explanation	Explain how text columns were converted
Train-test split explanation	Explain why data was divided
Underfitting model observation	Fill accuracy and behavior
Overfitting model observation	Fill accuracy and behavior
Good fit model observation	Fill accuracy and behavior
Final comparison table	Compare all three models
Business conclusion	Select the most suitable model

Important Note

Do not provide the final answer directly.

Learners should run the models, compare training and testing accuracy, and then identify whether the model is underfitting, overfitting, or giving a good fit.